Detector production pilot: formulas and results

NAUSICAA generated report

This is a pilot calculation using the recorded parameter cards. The EL variance ratio J is a provisional input for exercising the sampler; a calibrated $J(E/P)$ table is required for a physical production run.

Resolved inputs

Parameter	Value
TPC length	150.0 cm
Pressure	15.0 bar
Ion drift field	200.0 V/cm
Ion pitch, frame	3.0 mm, 30.0 ms
Electron hole pitch, bucket	10.0 mm, 2.0 μs
Electron lifetime	15.0 ms
EL E/P	3.00 kV/cm/bar
Pair-creation energy W, Fano factor F	21.90 eV, 0.150
Pilot EL variance ratio J	0.100

Calculations and assumptions

$$\mu_i = 10^6 E_i/W, \qquad \mathrm{Var}(N_i) \simeq F\mu_i,$$
$$L_e = L_{\max} - L_i, \qquad L_i \sim U(L_{\min}, L_{\max}),$$
$$\sigma_{i,T} = \sigma_{i,L} = 10\sqrt{2(k_B/e)TL_i/E_d}\ \mathrm{mm},$$
$$\sigma_{e,T/L} = d_{T/L}\sqrt{L_e/P}, \qquad p_{\mathrm{survive}} = e^{-L_e/(v_e\tau_e)},$$
$$Y_g = a_g \max(E/P - x_{0,g}, 0)Ph, \qquad \mathrm{Var}(N_{\mathrm{EL}}|n_e) = n_e J Y_g,$$
$$A = (h + r - \sqrt{h^2 + r^2})/(2h), \qquad N_{\mathrm{incident}}|N_{\mathrm{EL}} \sim \mathrm{Binomial}(N_{\mathrm{EL}}, A).$$

At each source hit, one integer pair draw creates both initial carrier populations. The minimum integer variance limits the Fano target at low occupancy. Electron attachment is sampled once per pair realisation and shared by the two gas alternatives. Diffused carriers enter physical grid cells directly; there is no diffusion-window charge loss. VUV photons touching a hole wall are absorbed; a circular SiPM covers its exit.

Derived quantity	Pure Xe	Xe–15% He
Max transverse electron σ (mm)	11.068	5.060
Max longitudinal electron σ (mm)	2.846	2.372
Mean EL yield (photons/electron)	2,332.4	2,293.5

Ion width at maximum drift: 1.947 mm. Ion frame: 3.0 mm. Electron bucket: 2.0 mm. Minimum electron survival: 0.9048. Direct optical acceptance: 0.19098. Gate/anode potentials relative to cathode: 30.0 / 52.5 kV.

Saved event totals and validation

Sample	Events	Pairs	Ions	Surviving e^-	Xe photons	Xe–He photons
bb0nu	3	336,748	336,748	312,961	139,416,209	137,073,762
single_electrons	3	336,601	336,601	328,125	146,167,619	143,715,327

The ion ledger is exact for both samples: ions in pad/time cells equal initial pairs. For each gas, incident photons do not exceed emitted photons. The three branch files store matching per-event pair checksums.

Provenance and scope

Run ID: `6c3af3e35b13b25c`. Code SHA-256 prefix: `55ff9d88b5518f51`. The complete digest is in the HDF5 summary. Peak process RSS: 841.0 MB. Source hit tables are the 1 mm Geant4 samples. Ion thermal diffusion follows ITACA Revisited, Eq. (2.1) (arXiv:2604.02443). The pure-Xe transverse coefficient is its Table 1 input; other diffusion coefficients are recorded model inputs. EL yield coefficients follow Fernandes et al., JHEP 04 (2020) 034, Table 1. The definition of J follows Oliveira et al., JINST 6 (2011) P05007, Eq. (4.6).